Question Paper

Exam Date & Time: 07-Jan-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

III SEMESTR B.TECH. (AI&ML ENGINEERING)
MAKE UP END SEMESTER EXAMINATION

LINEAR ALGEBRA AND LOGIC [MAT 2128]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions

1A) (4)

Find a generating set for the column space of the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 2 & 4 & 0 & -8 \\ 0 & 0 & 2 & 6 \end{bmatrix}.$$

Check whether the vector $u = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ is in the column space of A or not?

1B) (3)

Let $y = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $u = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$.

Write y as the sum of a vector in span $\{u\}$ and a vector orthogonal to the the vector 'u' .

1C) (3)

Obtain the principal disjunctive normal form and principal conjunctive normal form of

$$f(x_1, x_2, x_3) = (x_1 \wedge x_2) \vee (x_1 \wedge x_3) \vee (\overline{x_2} \wedge x_3).$$

2A) (4)

Suppose $T: R^2 \to R^2$ is a linear transformation such that $T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $T\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$.

a) Find $T\left(\begin{bmatrix} 3 \\ 3 \end{bmatrix}\right)$

b) Find $T(e_1)$

c) Find $T(e_2)$

d) Find $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$

2B) (3)

Give Cholesky's decomposition of

$$A = \begin{pmatrix} 4 & 10 & 8 \\ 10 & 26 & 26 \\ 8 & 26 & 61 \end{pmatrix}$$

2C) (3)

Show that $S \vee R$ is tautologically implied by

$$(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S).$$

3A) (4)

Find singular value decomposition of

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$$

3B) (3)

Determine the linear independency or dependency of the set of vectors

$$\left\{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right\}.$$

3C) (3)

Diagonalize the matrix

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

4A) (4)

Give the spectral decomposition of the matrix

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

4B) (3)

Let $b_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}, b_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}, c_1 = \begin{bmatrix} -7 \\ 9 \end{bmatrix}, b_2 = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$ and let the basis for R^2 is given by $B = \{b_1, b_2\}$ and $C = \{c_1, c_2\}$.

Find the change of co-ordinate matrix from C to B and hence find the change of co-ordinate matrix from B to C.

4C) (3)

Show that

$$(x)(P(x) \rightarrow Q(x)) \wedge (x)(Q(x) \rightarrow R(x)) \Rightarrow (x)(P(x) \rightarrow R(x))$$

5A) (4)

Let W be a subspace of R^5 spanned by $u = (1,2,3,-1,2)$ and $v = (2,4,7,2,-1)$.

Find a basis of the orthogonal complement $W^{\perp}$ of W.

5B) (3)

For all $u, w \in V$ where V is an inner product space, prove that

$$\|u + w\| \leq \|u\| + \|w\|.$$

5C) (3)

Make the change of variable that transforms the quadratic form
$Q(x) = 2x_1^2 - 4x_1x_2 - x_2^2$ in to a new quadratic form which has no cross-product terms.

-----End-----